

SBIX 1.0

SATYA BHARTI INNOVATEX

INNOVATION CHALLENGE

Sample Project

SMART URINE ANALYZER

An accessible and technology-based approach to health monitoring

SAMPLE PPT — FOR REFERENCE ONLY

This sample demonstrates the structure and level of detail. Participants must present their own original idea.

THINK • BUILD • INNOVATE

PROBLEM

The Problem We Want to Solve

Urine testing can help identify health-related conditions early, but access to testing is not always easy.

EXISTING CHALLENGES

- Testing usually requires visiting a laboratory.
- Testing can take time and may involve cost.
- Some rural and remote areas may have limited testing facilities.
- Because testing is not always convenient or accessible, people may delay checking potential health concerns.

WHY DOES THIS MATTER?

A simpler and more accessible approach could encourage people to monitor their health more regularly.

CURRENT JOURNEY

Person

Laboratory

Waiting

Result

TIME • ACCESS • COST



WHAT YOUR SLIDE SHOULD INCLUDE

- Real-world problem
- Who faces it
- Existing gaps
- Why it matters

SOLUTION

SMART URINE ANALYZER

A portable and easy-to-use device that analyzes a urine test strip using a camera and computer-vision-based colour analysis.

01 • COLLECT

A small urine sample is collected.

02 • TEST

The sample is passed over a urine analysis strip.

03 • CAPTURE

A webcam captures the strip after the reaction.

04 • ANALYZE

The captured colour is compared with a standard colour reference.

05 • DISPLAY

The system displays the result and may indicate possible health conditions or advise consultation with a doctor.

TARGET USERS

Individuals and community health settings

KEY IDEA

Detect earlier. Understand better. Act smarter.

SOLUTION FLOW

Sample



Test Strip



Camera



Analysis



Result

WHAT YOUR SLIDE SHOULD INCLUDE

- Proposed solution
- How it addresses the problem
- Target users
- Main purpose

What Makes Our Idea Different?

01 • CONTACTLESS & HYGIENIC

Designed around a hygienic testing process with camera-based analysis.

02 • COMPUTER VISION

A webcam captures the test-strip image and the system analyzes the colour.

03 • PORTABLE

Designed to make health monitoring more accessible beyond traditional laboratory settings.

04 • EASY TO USE

Aims to provide a simple process for obtaining and viewing results.

05 • EVERYDAY HEALTH MONITORING

Focuses on bringing health monitoring closer to everyday users.

KEY USP

PORTABLE + COMPUTER VISION + SIMPLE USER EXPERIENCE

WHAT YOUR SLIDE SHOULD INCLUDE

- What is new or creative?
- Unique features / USP
- What makes it different?
- Why it is innovative?

TECHNOLOGY STACK

Technologies Behind Our Solution

URINE ANALYSIS STRIP

Used for the testing process.

WEBCAM

Captures the test-strip image after the reaction.

PYTHON

Used for the software implementation.

STREAMLIT

Used to create the web-based interface.

COMPUTER VISION / COLOUR ANALYSIS

Compares the captured strip colour with a standard colour reference.

SYSTEM FLOW

URINE SAMPLE



ANALYSIS STRIP



WEBCAM



IMAGE / COLOUR ANALYSIS



PYTHON + STREAMLIT



RESULT DISPLAY

WHAT YOUR SLIDE SHOULD INCLUDE

- Hardware/components
- Software/tools
- Platforms/frameworks
- Purpose of each technology
- How they work together

HOW IT WORKS?

From Sample to Result

STEP 1 • SAMPLE

A small urine sample is collected.



STEP 2 • TEST STRIP

The sample passes over a urine analysis strip placed in the testing area.



STEP 3 • IMAGE CAPTURE

A webcam captures the strip after the reaction.

STEP 4 • COMPUTER VISION

The system analyzes the captured colour.



STEP 5 • COMPARISON

The detected colour is compared with a standard colour reference.



STEP 6 • RESULT

The result is displayed on the screen, indicating possible health conditions or suggesting consultation with a doctor.

WHAT YOUR SLIDE SHOULD INCLUDE

- Step-by-step working
- Input and output
- System/data flow
- Important components
- Flowchart / architecture

SAMPLE WORDING NOTE

Use "possible health conditions" and "consult a doctor" rather than claiming diagnosis.

INPUT → PROCESSING → ANALYSIS → OUTPUT

FEASIBILITY & SCALABILITY

FEASIBILITY

TECHNOLOGY

The prototype uses a camera-based analysis process with Python/Streamlit and a urine analysis strip.

PRACTICAL USE

The project is designed around a simple testing and result-display workflow.

ACCESSIBILITY

Its portable approach could be useful in settings where convenient access to testing is limited.

SCALABILITY

- More test parameters
- Improved computer-vision analysis
- Additional health-monitoring features
- Mobile connectivity
- Wider use in healthcare settings

EXPANSION PATH

PROTOTYPE

TEST &
IMPROVE

MORE
PARAMETER
S

MOBILE
CONNECTIVI
TY

WIDER
APPLICATIO
N

Start small → Test → Improve → Expand

WHAT YOUR SLIDE SHOULD INCLUDE

- Can it realistically be developed?
- Resources required
- Practical limitations
- How it can be expanded
- Who could use it?

FUTURE SCOPE

Where Can Smart Urine Analyzer Go Next?

01 • ADVANCED AI / ML

Develop AI/ML models for deeper analysis and broader disease/condition recognition.

02 • MOBILE APP CONNECTIVITY

Connect the system with a mobile application for real-time health tracking.

03 • CLOUD-BASED HEALTH HISTORY

Use cloud-based storage to maintain a user's health history.

FUTURE ROADMAP

**CURRENT
PROTOTYPE**



**ADVANCED
AI / ML**



**MOBILE
APP**



**CLOUD HEALTH
HISTORY**



**CONNECTED HEALTH
MONITORING**

A long-term vision: make health monitoring more connected, accessible and data-informed.

- Future improvements
- New features
- Expansion possibilities
- Long-term vision

Explain how the idea can be improved or expanded

CONCLUSION

Making Health Monitoring More Accessible

SAMPLE CONCLUSION

Smart Urine Analyzer proposes a technology-based approach to make urine-strip analysis more convenient and accessible.



TESTING STRIP



IMAGE CAPTURE



COMPUTER
VISION



DIGITAL
ANALYSIS

EXPECTED IMPACT

- Encourage regular health monitoring
- Make testing more convenient
- Support access in rural and remote settings
- Encourage people to seek professional medical advice when needed

- Problem summary
- Solution
- Main innovation
- Expected impact

THINK • BUILD • INNOVATE

THE JOURNEY

PROBLEM

Difficult access



IDEA

AI-assisted analysis



INNOVATION

Imaging + AI + interface



IMPACT

More accessible monitoring

“Innovation begins when technology is used to solve a real-world problem.”

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Bharti Airtel
Foundation



THANK YOU

